

- 5 Fig. 5.1 below shows an isolated, metal sphere in a region of vacuum that carries a negative electric charge.

E-field lines: directed radially towards centre of sphere, equally spaced around sphere, giving a symmetrical pattern. There should be no field lines inside sphere.

Equipotential lines: concentric circles perpendicular to field lines, spacing is further apart with distance from centre

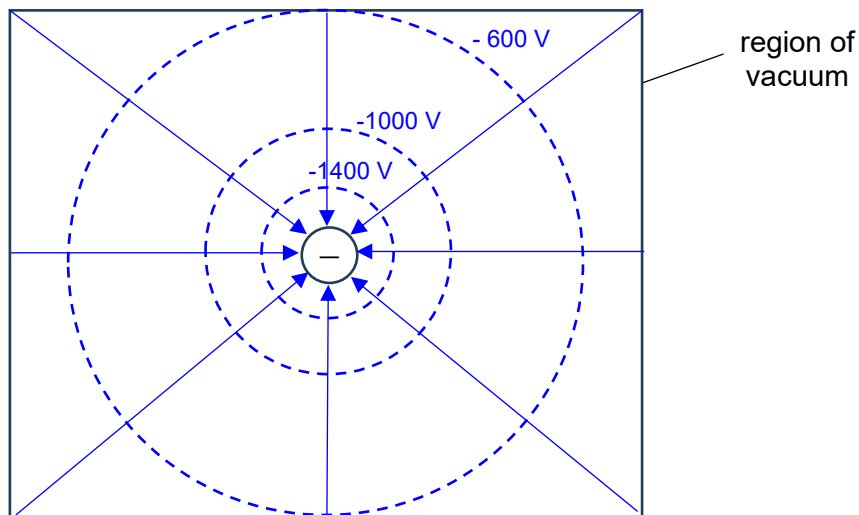


Fig. 5.1 (not to scale)

- (a) The electric potential at the surface of the sphere is -1800 V . In the region of vacuum on Fig. 5.1, draw
- (i) arrows to represent the electric field pattern outside the sphere [1]
 - (ii) dotted lines to represent three equipotential surfaces of -1400 V , -1000 V and -600 V outside the sphere. Label the potentials clearly. [1]

- (b) On the axes given in Fig. 5.2, sketch a graph to show the variation with distance r from the centre of the sphere of the potential V . The dotted line is drawn at $r = R$ where R is the radius of the sphere.

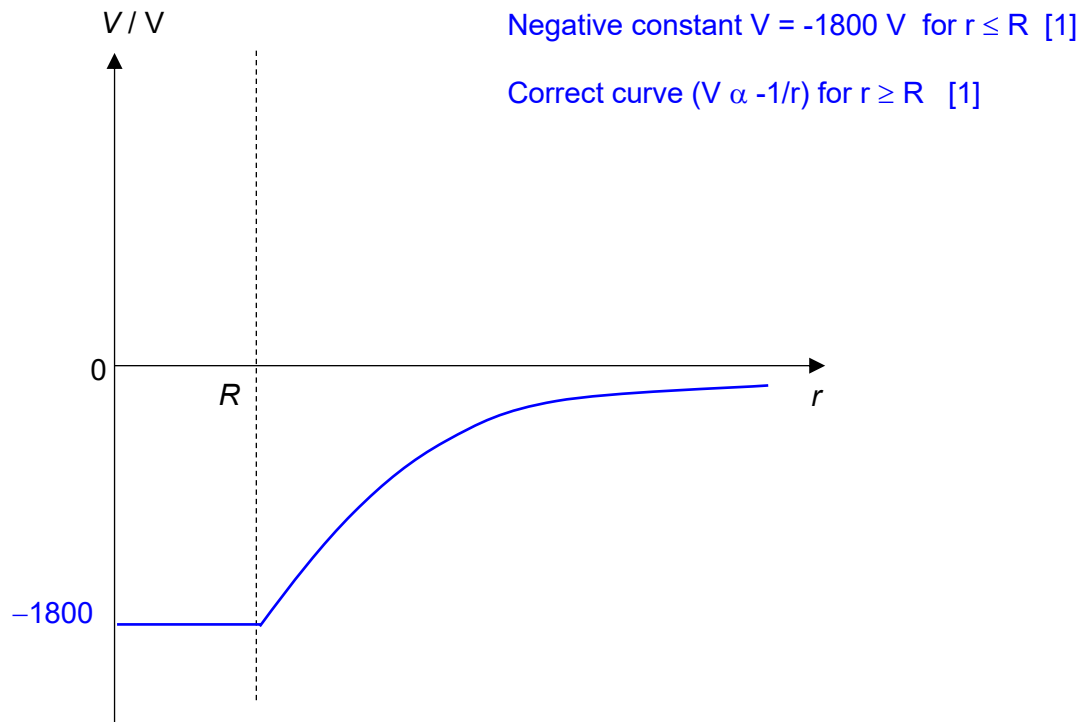


Fig. 5.2

[2]

- (c) The sphere carries an electric charge of -9.0 nC and has a radius of 4.5 cm . An electron is initially at rest at the surface of the sphere.

- (i) Describe the motion and path followed by the electron as it leaves the surface of the sphere.

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 Electron accelerates or speed increases [1]
 as it moves radially outwards/perpendicular to surface of sphere (along a field line) [1]

 [2]

- (ii) Determine the speed of the electron when it reaches a point a distance 0.30 m from the centre of the sphere.

By conservation of energy,

Loss in EPE = gain in KE

$$q [Q/(4\pi\epsilon_0 (0.045)) - Q/(4\pi\epsilon_0 (0.30))] = \frac{1}{2} m v^2 - 0 \quad \text{or} \quad q [(-1800) - Q/(4\pi\epsilon (0.30))] = \frac{1}{2} m v^2 - 0$$

$$(-1.6 \times 10^{-19}) \times (-9.0 \times 10^{-9}) \times [1/(4\pi\epsilon (0.045)) - 1/(4\pi\epsilon (0.30))] = \frac{1}{2} (9.11 \times 10^{-31}) v^2 - 0 \quad [1]$$

$$v = 2.3 \times 10^7 \text{ ms}^{-1} \quad [1]$$

speed = m s^{-1} [2]

[Total: 8]

